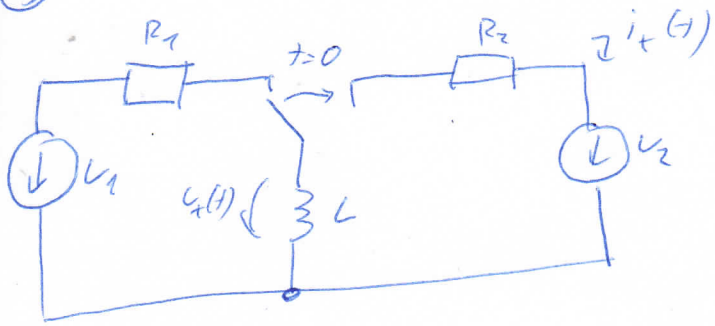
③



$\cancel{i_x(0)=0}$

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$V_1 = 5V$

$V_2 = 10V$

$L = 2H$

$R_1 = 1\,k\Omega$

$R_2 = 2\,k\Omega$

$i_x(0) = 0$

$$V_2 = i_x R_2 + L\frac{di_x}{dt}$$

$$\frac{V_2}{R_2} = i_x + \frac{L}{R_2}\frac{di_x}{dt} \rightarrow 0 = 1 + \frac{L}{R_2}\lambda$$

$$\lambda = \frac{-R_2}{L}$$

$$i_x(t) = K_1 e^{-\frac{R_2}{L}t} + k_2 = k_1 e^{\frac{-t}{\tau}} + k_2 \quad \tau = \frac{L}{R_2}$$

$$k_2 = \frac{V_2}{R_2} = 0{,}005$$

$$0 = k_1 e^{\frac{0}{\tau}} + 0{,}005 \mapsto k_1 + 0{,}005 = 0 \mapsto k_1 = -0{,}005$$

$$i_x(t) = -5 e^{\frac{-t}{\tau}} + 5\cdot 10^{-3} \quad \tau = \frac{2}{2000} = 0{,}001$$

$$\boxed{i_x(t) = -5\cdot 10^{-3} e^{\frac{-t}{10^{-3}}} + 5\cdot 10^{-3} \ [A]}$$